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MODELING OF CYBER-PHYSICAL SYSTEMS AND DIGITAL TWIN BASED ON EDGE COMPUTING, FOG COMPUTING AND CLOUD COMPUTING TOWARDS SMART MANUFACTURING

Qinglin Qi,
School of Automation Science and Electrical Engineering, Beihang University (BUAA)
Beijing, 100083, China
Email: kylin3366@buaa.edu.cn

Dongming Zhao,
Department of Electrical and Computer Engineering, University of Michigan-Dearborn
Dearborn, Michigan, USA
Email: dmzhao@umich.edu

T. Warren Liao,
Department of Mechanical and Industrial Engineering, Louisiana State University, Baton Rouge, LA 70803, USA
Email: ieliao@lsu.edu

Fei Tao*
School of Automation Science and Electrical Engineering, Beihang University (BUAA)
Beijing, 100083, China
Email: ftao@buaa.edu.cn

ABSTRACT

Nowadays, smart manufacturing has attracted more and more interesting and attentions of researchers. As an important prerequisite for smart manufacturing, the cyber-physical integration of manufacturing is becoming more and more important. Cyber-physical systems (CPS) and digital twin (DT) are the preferred means to achieve the interoperability and integration between the physical and cyber worlds. From the perspective of hierarchy, CPS and DT can be divided into unit level, system level, and SoS (system of system) level. To meet the different requirements of each level, the following three complementary technologies, i.e., edge computing, fog computing and cloud computing, are instrumental to accelerate the development of various CPS and DT. In this article, the perspectives of unit-level, system-level, and SoS-level of CPS and DT supported by edge computing, fog computing and cloud computing are discussed.

Keywords: Cyber-physical system (CPS), digital twin (DT), edge computing, fog computing, cloud computing, smart manufacturing.

INTRODUCTION

The advances in new generation information technologies (New IT), such as Internet of Things (IoT), big data, cloud computing, artificial intelligence (AI), etc., have had a profound impact on manufacturing. In this context, many countries have issued their advanced manufacturing strategies, e.g., Industry 4.0 in Germany, Industrial Internet in USA, Made in China 2025, etc. Although each of these strategies was proposed under different circumstances, one of the common purposes of these strategies is to achieve smart manufacturing [1]. However, to implement smart manufacturing, one specific challenge is how to converge the physical and cyber worlds of manufacturing [2]. With the initiatives of various advanced manufacturing strategies, cyber-physical integration has gained extensive attentions from academia, industry, and the government.

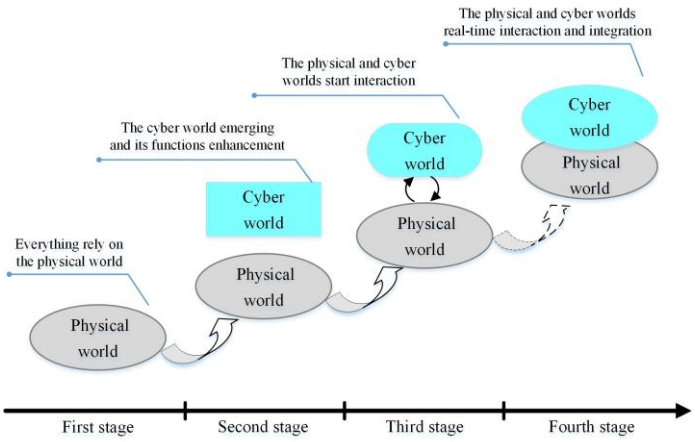


Figure 1 The evolution process of manufacturing [2].

From the perspective of the manufacturing development, a fusion process between the physical and cyber worlds of manufacturing is evolving over time, as shown in Figure 1 [2]. Firstly, before the advent of information technologies, manufacturing only involved the physical world. Specifically, all manufacturing activities are done by human and/or machines. The lack of information limited the manufacturing efficiency and production capacities. In the second stage, the cyber world has appeared. A variety of manufacturing-related software is developed and applied in production. However, due

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to the technological limitation, the advantage and potential of cyber world had not been fully exploited. In the third stage, the physical world and the cyber world began to interact with each other. The cyber world strengthened the management of the manufacturing process, optimizing the logistics, capital flow, and information flow. However, during this stage, the physical and cyber worlds of manufacturing are not synchronized. At present, with the continuous developments of New IT, the real-time interaction and further integration between the physical and cyber worlds of manufacturing is becoming the inevitable trend towards smart manufacturing.

According to either the horizontal analysis from various manufacturing strategies, or vertical analysis from manufacturing development, the cyber-physical integration of manufacturing is becoming more and more important. To this end, cyber-physical systems (CPS) and digital twin (DT) are the preferred technologies. Integrating computing, communication and control capabilities with the dynamics of physical systems, CPS offers close interaction between cyber and physical components [3]. Digital twin is to create the virtual models for physical objects in the digital way to simulate their behaviors, serving as a bridge between the physical and the cyber worlds [4]. Through the cyber-physical closed loop, CPS and DT could promote the smartness of all manufacturing processes.

Smart manufacturing is the value creation process, in which multiple subjects participate. Depending upon different magnitude of the data in smart manufacturing, and different time-sensitivity of smart applications, CPS and DT can be divided into three different levels, i.e., the unit level, system level, and SoS (system of system) level [5]. Each level has the data closed-loop, including status perception, data analysis, decision making, and execution. However, as various levels have different roles and infrastructures, how to build various levels of CPS and DT towards smart manufacturing becomes a central issue.

Edge computing, fog computing and cloud computing may offer solutions to the above-mentioned issue. Cloud computing enables ubiquitous, convenient, on-demand network access to the shared resources pool [6]. Because of the on-demand resources sharing, high computational and storage capabilities, as well as low costs, cloud computing is a perfect technological means for SoS-level CPS and DT. Fog computing shifts the compute, storage, and networking capabilities of cloud to the edge network [7]. By directly processing data on the network (e.g., network routers, various information systems, etc.), fog computing could help implement system-level CPS and DT. Edge computing allows data processing to be performed at closer proximity to the data sources [8]. The edge can be defined as the end points with both data producers and data consumers included. For unit-level CPS and DT, edge computing would be very beneficial.

Therefore, edge computing, fog computing and cloud computing are introduced as technologies for implementing the unit-level, system-level and SoS-level of CPS and DT, respectively, towards smart manufacturing. Next, a solution framework based on edge computing, fog computing and cloud computing is provided.

CYBER-PHYSICAL SYSTEMS (CPS) AND DIGITAL TWIN (DT) IN MANUFACTURING

The notion of integration between the physical and cyber worlds is increasingly embraced by manufacturers. Manufacturing is no longer the simple standalone physical machining process, but the interactive process of the physical and cyber worlds by mutual feedback. The CPS and DT provide technical basis for the cyber-physical fusion. By integrating advanced information, communication and automatic control technologies, CPS and DT enable mutual mapping, timely interaction and efficient coordination between the physical and cyber worlds.

A. CPS in manufacturing

As the next-generation smart systems, CPS integrate computing, communication, control, and other technologies. The powerful computing and communicating capabilities in the cyber world are used to monitor, control, and coordinate the operations of engineered systems in the physical world [9]. The seamless integration and interaction of computational elements and physical components is the key to CPS. With higher degree of interconnectivity and feedback, computational entities can affect physical processes and vice versa, which are in intensive communication with each other [10]. In this way, CPS enable the cyber world to control the physical entities in the physical world in a remote, reliable, real-time, safe, collaborative, robust, and efficient way [11].

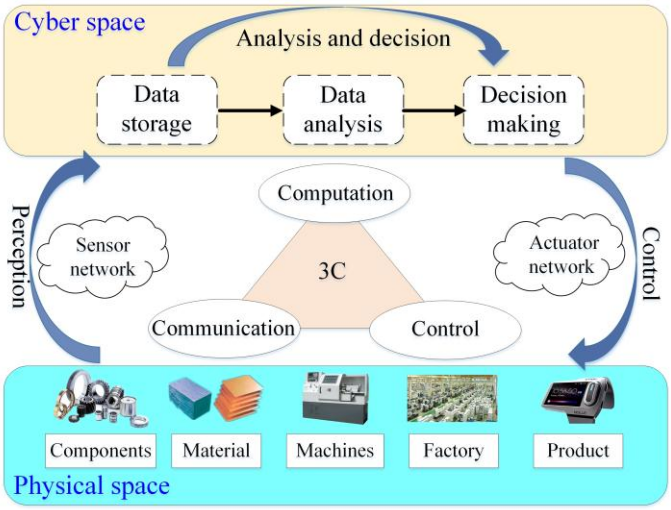


Figure 2 The CPS in manufacturing.

Due to the potential benefits to society, economy and environment, CPS has gained extensive attentions in various fields, especially manufacturing [12, 13]. In the manufacturing environment, CPS are increasingly used as networked processing elements to control physical processes or devices, and can be characterized by smartness, connectedness and responsiveness towards internal and external changes [14]. As shown in Figure 2, CPS in manufacturing involves both the cyber world and the physical world. On the basis of perception based on sensor network, the physical objects (e.g., material,

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machines, factory, products, etc.) are connected to the cyber world. The cyber world acquires and processes large amounts of data from the physical world. Through data analysis and decision making, these data are devoted to a variety of network-based applications. Ultimately, these applications would adjust the physical world through the actuators. As a result, the manufacturing system has the functions of communication, precise control, remote coordination and autonomy. In the process of interaction between the cyber and physical worlds of manufacturing, many smart applications are achieved.

In order to promote the CPS and their applications in the manufacturing field, many researchers have devoted to it. A 5-level CPS architecture was proposed as a guideline for the implementation of CPS [15]. Leitão et al. [16] described four prototype implementations for industrial automation based on CPS. Wang et al. [17] presented an approach for CPS implementation using the combined strength of holons, agents and function blocks. CPS can also be implemented in the cloud environment for the cloud-enabled prognosis, remote accessibility and controllability of machines and processes across spatial boundaries. The current state-of-the-art and the latest advancement of CPS in manufacturing is presented in [18] together with directions of future research and applications in the field. However, little of the existing research has considered the different granularity of CPS as the basic module of resource organization.

B. DT in manufacturing

In the general manufacturing process, designers, manufacturers, and after-sales serviceman are affiliated with different departments or systems, and work independently. Therefore, how to feed back the effective information collected from manufacturing process to the product design link, how to realize the correlation and dynamic adjustment between manufacturing planning and implementation, how to promote proactive MRO in digital way, are big challenges. Digital twin paves a way to solve the above-mentioned challenges. Digital twin can allow companies or users to have a complete digital footprint of product from design to the end of product life cycle.

The concept of digital twin was firstly presented by Grieves at one of his presentation about PLM in 2003 [19]. Taking previous definitions into account, in our opinion digital twin is not just a mapping of static physical systems, but shall also be a dynamic simulation of physical systems. It contains five parts, i.e. services, physical entities, virtual models, the connected data, and the interaction and connection between them. As shown in Figure 3, digital twin reflects two-way dynamic mapping of physical objects and virtual models [20]. Virtual models of physical objects are created in a digital way to simulate their behaviors in real-world environments [21]. On the other hand, the virtual models could guide the physical entities to respond to manufacturing changes, and to improve the operation [20]. Besides, services with their characteristics of interoperability and platform independence pave the way for large-scale smart applications and collaborations. Through services, DT realizes the closed-loop optimization in the entire process from product design, manufacturing execution, to predictive maintenance [22]. In the process of interaction between virtual, reality and services, data integration is an inevitable trend.

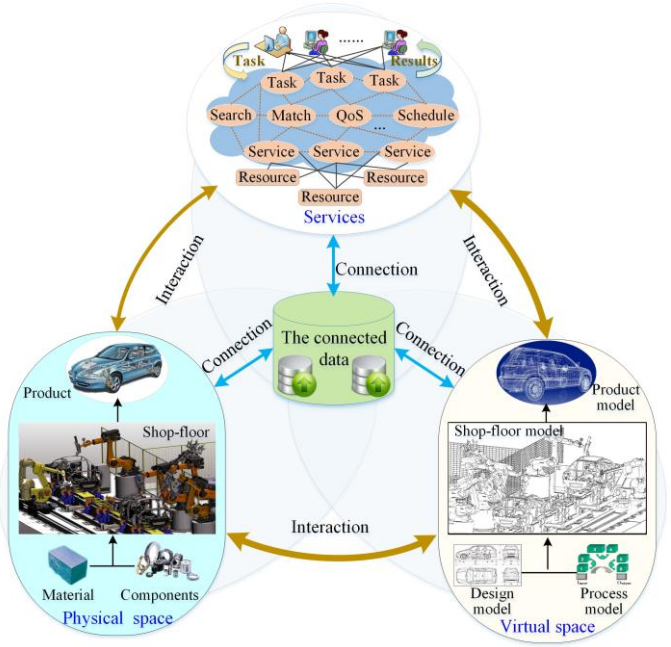


Figure 3 The digital twin in manufacturing.

As an effective means to achieve the cyber-physical integration, digital twin attracts more and more attentions. For example, the digital twin is used to predict the reengineering aircraft structural life [23]. Seshadri and Krishnamurthy demonstrated the great advance in predicting the damage location, size and orientation [24]. Nowadays, as the application of New IT in manufacturing, the smart manufacturing era is coming and digital twin has also been introduced into manufacturing stage. For instance, the concept of digital twin shop-floor (DTS) [2] was proposed. Vachálek et al. investigated the application of digital twin in an industrial production line and analyzed its superiority [25]. Furthermore, Gabor et al yielded an architectural framework of the digital twin for complex systems to optimize the system behaviors [26]. It is worthwhile to note that the above-mentioned digital twin applications focus primarily on production and/or product, and do not consider the different granularity of DT.

THE HIERARCHICAL LEVELS OF CPS AND DIGITAL TWIN IN MANUFACTURING

CPS and DT are the core supporting technologies of smart manufacturing. To better understand CPS and DT, a systematic and hierarchical view should be established. From the perspective of hierarchy, CPS and DT can be divided into three different levels, which are unit level, system level, and SoS (system of system) level [5], as shown in Figure 4.

The unit level CPS is the smallest granularity with indivisibility. Some small units, such as a component and equipment (e.g., NC machine tools, smart robot, etc.), material (e.g., raw material with RFID, Automated Guided Vehicle with

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built-in sensors, etc.) and environmental sensors, can form the unit-level of CPS. For example, smart NC machine tools can utilize knowledge to make decisions according to the spindle load changes. Smart NC machine tools can adjust the feed rate, such as increasing the feed rate in the case of smaller loads, and reducing the feed rate in the case of larger loads, to shorten the processing cycle, improve processing efficiency and quality. With respect to digital twin, the unit-level DT (e.g., component, material, equipment, etc.) can also be formed through the description and modeling in the virtual space of geometric shape, identity and function information, as well as operating status of unit-level physical entities. Accompanying the physical machining, assembly, and integration, both the virtual and physical spaces of unit-level DT evolve in the process of virtual-reality interaction. Compared with CPS, digital twin pays more attention to the construction of the models, including geometric shape model, rules, behavior and other constraints models, etc. At the unit level, the cyber space of CPS can control the physical hardware, whereas the virtual space of DT can further perform high-fidelity visual simulation.

System level CPS is the integration of multiple unit-level CPSs, which cooperate with each other. Through the industrial network (such as industrial field bus, Industrial Ethernet, etc.), multiple unit-level CPSs achieve a wider range of system-level CPS. For example, a smart production line composed by NC machine tools, smart robots, AGV car, and conveyor belt, etc. is a system-level CPS. A smart shop-floor with the functions of smart production, which consist of "human-machine-material-environment", is also a system-level CPS. Based on the closed-loop of "perception-analysis-decision-execution" of multiple unit-level CPSs, the system-level CPS can improve the optimal allocation of manufacturing resources, and the collaboration efficiency among various resources. As for digital twin, system-level physical manufacturing system (e.g., production line, shop floor, factory, etc.) is same as the system-level CPS. Different from system-level CPS, the virtual models of unit-level DTs need to be combined and collaborate with each other to form a system-level virtual model. In the interaction of the unit-level virtual models, the optimal organization of physical resources may be found. Besides, a complex product may be considered as a system-level DT. For example, for an aircraft consists of massive components and parts, the engine DT is used to assess the running state and help find the subsequent damage in a real-time way. The wing DT is used to simulate the flight attitude. These correlated components and parts DTs in combination form a complex product DT.

By building a CPS smart service platform, collaborative optimization between system-level CPSs can be achieved. At this level, multiple system-level CPSs constitute the SoS-level CPS. For example, the collaboration between multiple production lines or multiple factories, enables the integration of the whole product lifecycle processes. Through the smart service platform, the cross-systems and cross-platforms interconnection and interoperability can be achieved. For instance, the enterprises collaboration networks derived from complex network-based manufacturing services supply-demand matching [27] would provide different kinds of cooperation applications, such as commerce cooperation, supply chain cooperation, manufacturing cooperation, etc. Besides, based on SoS-level CPS, some applications involved multiple participants, such as personalized customizations, smart design, remote maintenance, etc., can be achieved. As for SoS-level DT, it is not simply composed of system-level DTs. The SoS-level DT is the integration of the various stages of the product lifecycle. It brings the data from all aspects of product lifecycle together, hence laying the foundation for innovative product design and quality traceability. Besides, the SoS-level DT promotes the efficient synergies between the different stages of product lifecycle. For example, the data from manufacturing link and maintenance link can also help improve DT based design. SoS-level DT can not only shorten the design cycle but also greatly reduce the cost of time and money.

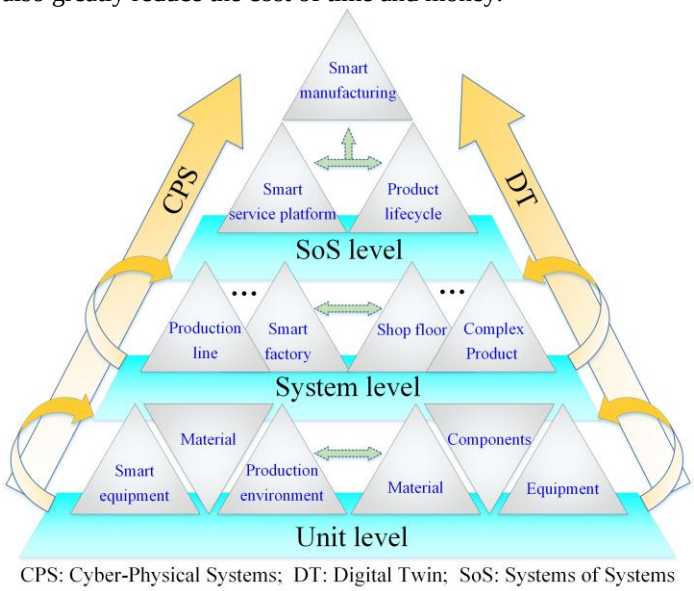


Figure 4 The hierarchical levels of CPS and DT in manufacturing.

MODELING OF VARIOUS LEVELS OF CPS AND DT BASED ON EDGE COMPUTING, FOG COMPUTING AND CLOUD COMPUTING

In light of the widespread applications of New IT in manufacturing, manufacturers are facing the explosive growth of data [28]. The essence of CPS and DT is to acquire data from physical entities and environment thorough sensors, as well as compute and analyze them in the cyber world, so as to control the physical entities and environment. As a result, the data closed-loop is established, forming the interaction and fusion between the physical world and the cyber world. The various levels of CPS and DT in manufacturing have different requirements for data processing and data circulation, such as latency, bandwidth, security, and so on. The edge computing, fog computing and cloud computing with complementary properties, provide new ideas and way for the implementation of the unit-level, system-level and SoS-level CPS and DT.

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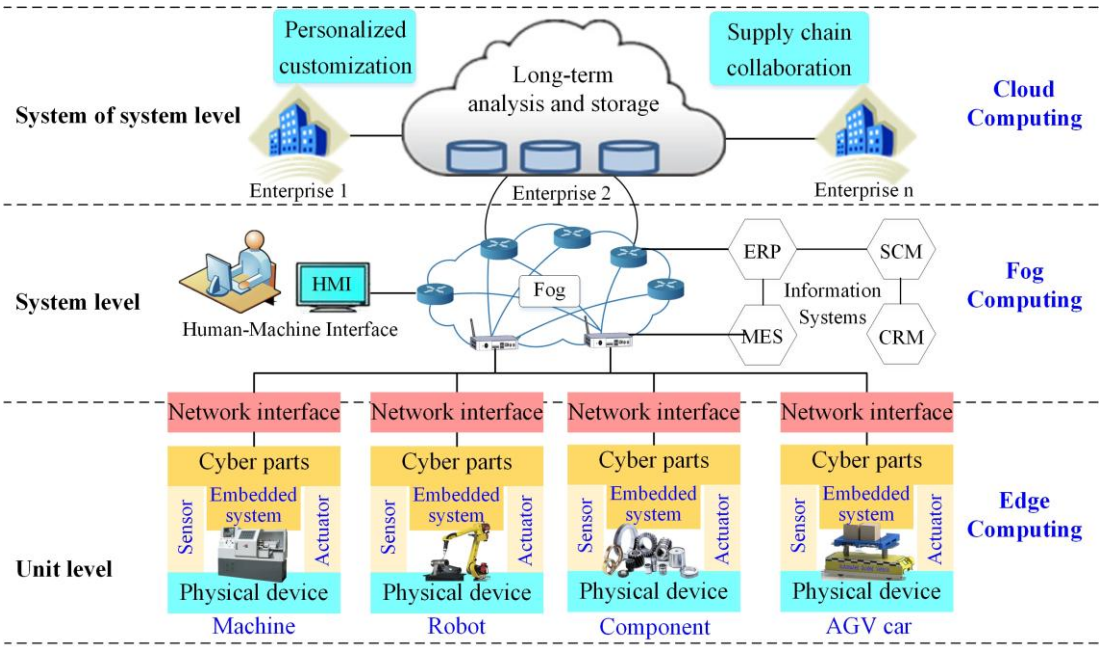


Figure 5 Modeling of various levels CPS and DT in manufacturing based on edge computing, fog computing and cloud computing

A. Cloud computing, fog computing and edge computing

Cloud computing is the computing paradigm that enables ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., computing and storage facilities, applications, services, etc.) [6]. Through virtualization technology, cloud computing shields the diversity of underlying devices and provides users with a variety of services in a transparent way, including IaaS (Infrastructure-as-a-Service), PaaS (Platform-as-a-Service) and SaaS (Software-as-a-Service) [29]. Due to the increasing number of access devices, cloud computing may face some problems in the bandwidth, latency, network unavailability, security and privacy, etc. Fog computing is considered as an extension of cloud computing to the edge network, providing services (e.g., compute, storage, networking, etc.) closer to near-user devices (e.g., network routers, various information systems, etc.), instead of sending data to cloud [7]. In fog computing paradigm, data storage and processing rely more on local devices, rather than on cloud data center. Fog computing makes the applications more convenient, meeting a wider range of node access. Similar to fog computing, edge computing also allows computation to be performed at the edge of the network, but at closer proximity to the data sources [8]. The difference between fog computing and edge computing is that fog computing relies on interconnection capabilities among nodes, whereas edge computing runs in isolated edge nodes. Edge computing provides edge services near the source of data to meet the critical requirements in agile connectivity, real-time optimization, smart applications, security and privacy.

B. Modeling of various levels of CPS and DT based on edge computing, fog computing and cloud computing

The basic requirements that need to be met in constructing a unit-level CPS and DT include the following: 1) state perception; 2) compute and process data; 3) physical entities control. As an architecture that extends computing, networking and storage capabilities from the cloud to the edge, edge computing enables perception, computing, and control for objects through data analysis and processing in edge nodes. As shown in Figure 5, the manufacturing resources (e.g., machine, robot, component, AGV car, etc.) with physical device (e.g., machine body, spindle, tools, etc.) and cyber part (e.g., embedded system), form the unit-level CPS and DT. Through the sensors, the cyber part can monitor and perceive information from physical devices, and exert control onto the physical devices through the actuator capable of receiving control instructions. Due to the perception, data analysis and control capacities, edge computing could be deployed on the unit-level CPS and DT, which can be considered as an edge node. Because of the data circulating on the unit device, the edge computing could implement smaller applications that can help provide more real-time responses. For example, on a machine tool or robot arm, sensors are used to detect whether some malfunction pose safety hazard or not. In this case, any delay is not tolerable. If the data is sent to cloud, the response time may be too long. However, with edge computing the time delay is reduced allowing decisions to be made in real-time because data is not far from the data source. Edge computing architecture does not rely on the Internet connection, which is beneficial to unit-level CPS and DT.

As shown in Figure 5, multiple unit-level CPSs and DTs are connected to the network through the network interface and the information management systems (e.g., ERP, MES, SCM, CRM, etc.). The system-level CPS and DT integrate a variety of heterogeneous unit-level CPSs and DTs. Besides, through the human-computer interface (HMI), each unit-level CPS and DT can be accessed and controlled to monitor and diagnose the

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state and health of the corresponding system. System-level CPS and DT emphasize the interconnection and interoperability among its component elements. On this basis, it focuses on the real-time and dynamic collaborative control of different elements, to achieve the coordination and unification of the physical world and the cyber world. In general, system-level CPS and DT are geographically concentrated (mainly within the manufacturing enterprise), which is well suited to the fog computing model. The data of system-level CPS and DT (i.e., enterprise internal data) can be directly dealt with fog computing to improve efficiency, rather than spread to the cloud and back again from the cloud. Taking delays, network traffic, cost, etc. into account, fog computing provides services at the edge of network, which promotes real-time interaction, scalability and interoperability. The fog computing environment could be constituted by the network components (such as, routers, proxy servers, base stations and others). These components can provide different computing, storage, networking capabilities to support the implementation of smart production applications. The unit-level CPS and DT use sensors to collect operating data on each device and analyze them. The fog computing environment summarize the data from all unit-level CPSs and DTs, and provide actionable information (such as cooperation with each other to finish task) for smart production. For example, in the actual production process, the robot places the material or parts on the conveyor belt or AGV for transportation, and then machine tools machine them into the desired product. In this way, the combination of real-time edge computing and fog computing with interoperability and scalability, would provide the best performance for system-level CPSs (e.g., production lines, shop floor, factory, etc.).

On the basis of system-level CPS and DT, it is able to achieve the collaborative optimization between system-level CPSs and DTs, by developing the smart service platform. As a result, multiple system-level CPSs and DTs constitute SoS-level CPS and DT. For example, multiple production lines or factories collaborate with each other through smart cloud services platform, achieving the enterprise-wide system integration of the whole product lifecycle. SoS-level CPS and DT involve various participants and resources, which may be geographically dispersed. The data of SoS-level CPS and DT is richer and more diverse. Therefore, the requirements that need to be met in SoS-level CPS and DT include 1) distributed data storage and processing, 2) provide data and smart services for enterprise collaboration. As shown in Figure 5, cloud computing architecture is conducive to the organization and management of a large number of connected devices, as well as the combination and integration of enterprise's internal and external data. In the cloud computing architecture, a variety of different types of storage devices can work together through the application software, to jointly provide data storage and business access for enterprises. Besides, massive data mining must be supported by the distributed processing, and virtualization technologies, which are the typical characteristics of cloud computing. Therefore, cloud computing is the ideal technology for long-term and massive data storage and analysis of SoS-level CPS and DT. In addition, services with the characteristics of interoperability and platform independence, pave the way for collaboration among system-level CPSs and DTs (e.g., factories or enterprises, etc.). In virtue of the cloud architecture, various unit CPSs and DTs, or system-level CPSs and DTs could be encapsulated into services, so as to become plug-and-play components and be shared by other participants [30]. In this way, manufacturers not only could conveniently provide their services to their collaborative stakeholders, but also could use various resources in a pay-as-you-go manner [31]. In summary, the cloud computing architecture provides more architectural flexibility and utilization of external data for the innovation of product and value creation. For example, supply chain applications and product customization can benefit from understanding the aggregate data view.

SUMMARY AND FUTURE WORKS

Regarding the development of manufacturing servitization, socialization, and demand personalization, the promotion and applications of smart manufacturing are the inevitable trend. Achieving the interaction and integration of the physical world and the cyber world of manufacturing through the use of the state-of-the-art technologies, constitutes an important prerequisite for implementing smart manufacturing. To this end, CPS and DT are the preferred technologies. CPS and DT can be applied to the entire manufacturing process, the whole supply chain, and the whole product lifecycle. Depending on the scope of resource allocation, CPS and DT can be stratified to unit-level, system-level, and SoS-level. Because the data generated in a different level has different requirements and uses at time scales, edge computing, fog computing and cloud computing offer ideal technical solutions for different-level CPS and DT.

Edge computing, fog computing and cloud computing cooperate with each other, better meeting the requirements of each level CPS and DT. However, there still are the following works to be undertaken and improved with the development of New IT. 1) More edge nodes need to be built and applied in manufacturing resources, and the capacities of edge computing need to be further improved. 2) The architecture, platform, and standard of fog computing are more attractive for end users. 3) The data processed by edge computing, fog computing and cloud computing is different. Thus, the algorithms and rules for data filtering between different levels are crucial. 4) More attention should also be paid to the issues of security and energy consumption minimization.

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